

Attack Mode

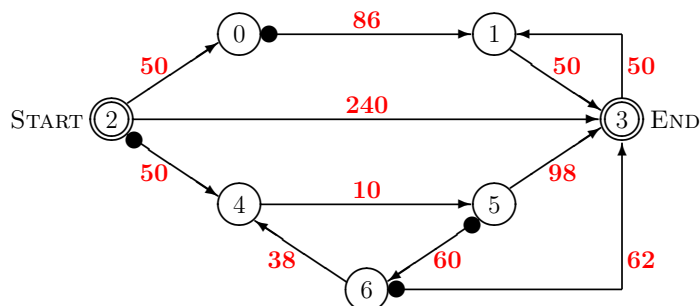
In 2026, there will be a new FIA race for electric cars, whose rules are inspired by those of Formula E. The race takes place in a city but, instead of comprising several laps in a circuit, it is up to the drivers to choose the route from the start to the finish line. Of course, the winner is whoever reaches the finish line first. Additionally, all cars must have the same characteristics and can drive in *attack mode*. When attack mode is activated, the car gains power and its speed doubles. Attack mode can only be switched on at the beginning of some streets and its effect lasts until the end of the street. This feature can be activated a given number of times, called the *maximum number of attacks*.



Designed by brgfx / Freepik

Ada is eager to participate in the race. As soon as FIA announced the city where it will take place, specifying the start and the finish squares, the streets that can be used (and their directions), the streets in which attack mode can be activated, and the maximum number of attacks, Ada estimated how long it would take her to drive through each street (with attack mode turned off). But, as there are many paths from the start to the finish line, she cannot figure out which are the *best strategies*, that is, which routes she should choose and where attack mode should be activated in order to reach the finish line in the least possible time. Can you help her?

Suppose the race takes place in a city area whose map is depicted below, comprising 7 squares connected by 11 streets. The race starts at square 2 and ends at square 3. An arrow from a square q_1 to a square q_2 means that there is a one-way street from q_1 to q_2 . Its label is Ada's estimated time, in *race time units* (RTU), to drive through the street with attack mode turned off. Streets where attack mode can be activated have a circle at the arrow's tail.



If the maximum number of attacks was two, the best strategy would be to choose route 2, 4, 5, 6, 3, activating attack mode twice: at the street from square 5 to square 6 and at the street from square 6 to square 3. The *estimated duration* of this strategy is 121 RTU ($121 = 50 + 10 + \frac{60}{2} + \frac{62}{2}$).

Task

Write a program that, given the specifications of the city area where the race will take place, the maximum number of attacks, and Ada's estimates, computes the estimated

duration of the best strategies. It is guaranteed that, for the given inputs, there is some route from the start to the finish line.

Input

The input first line has four integers, Q , S , A and T , which denote, respectively, the number of squares, the number of streets, the maximum number of attacks and the number of test cases. Squares are identified by integers, ranging from 0 to $Q - 1$.

Each of the following S lines contains four integers, q_1 , q_2 , t and m , which indicate that there is a one-way street from square q_1 to square q_2 which Ada estimates she can drive through in t RTU, when attack mode is turned off. Assume that $q_1 \neq q_2$ and t is even (i.e., $t = 2k$, for some positive integer k). The last number m is 1 or 0, depending on whether attack mode can or cannot be activated.

T lines follow, each one with two integers, s_i and e_i , specifying that the race starts at square s_i and ends at square e_i in the i^{th} test case (for every $i = 1, \dots, T$).

Constraints

- $2 \leq Q \leq 5\,000$ Number of squares
- $1 \leq S \leq 25\,000$ Number of streets
- $1 \leq A \leq 5$ Maximum number of attacks
- $1 \leq T \leq 6$ Number of test cases
- $2 \leq t \leq 10\,000$ Estimated time to drive through a street (in RTU)

Output

The output consists of T lines, each one with a single integer. The number on the i^{th} line represents the estimated duration (in RTU) of the best strategies in the i^{th} test case (for every $i = 1, \dots, T$).

Sample Input

```
7 11 2 3
2 0 50 0
2 3 240 0
2 4 50 1
0 1 86 1
1 3 50 0
3 1 50 0
4 5 10 0
6 4 38 0
5 6 60 1
6 3 62 1
5 3 98 0
2 3
6 4
2 1
```

Sample Output

```
121
38
93
```